

1. What is a Graphics object?

1. The Graphics object represents the part of the Frame that you can draw on.
2. The Graphics object represents the entire monitor.
3. The Graphics object represents the whole Frame.
4. The Graphics object represents the graphics board.

2. What method must a button listener implement?

1. actionPerformed().
2. buttonAction().
3. eventAction().

4. actionPerformed().

3. In Java, what do you call an area on the screen that has nice borders and various buttons along the top border?

1. A box.
2. A window.
3. A screen.
4. A frame.

4. What method changes the color of a content pane?

1. setBackground()
2. setBackgroundColor()
3. setForegroundColor()
4. setColor()

5. How should a command string be compared with a literal string?

1. equals()
2. =
3. equalsIgnoreCase()
4. ==

6. How can the user be prevented from entering text into a text field?

1. By using setVisible(false)
2. By using setEditable(false)
3. By using setEditable(true)
4. By using setNotEditable()

7. When is the paint() method of a frame object called?

1. The Java system calls it once when the program starts.
2. The main() method calls it once when the program starts.
3. The user calls it to display the frame.
4. The Java system calls it every time it decides to display the frame.

8. When the user clicks on a button, what is generated?

1. A Button object.
2. A Container.
3. An Event.
4. A Listener.

9. What type of object determines where GUI components go in the content pane?

1. The layout manager.
2. The layout programmer.
3. The frame hanger.
4. The layer organizer.

10. Which of the following are correct event handling methods?

1. `MousePressed(MouseClick e){}`
2. `mousePressed(MouseEvent e){}`
3. `functionKey(KeyPress k){}`
4. `componentAdded(ContainerEvent e){}`

11. The size of a frame on the screen is measured in:

1. nits
2. dots
3. inches
4. pixels

12. What is the name for a method that responds to events?

1. An application method.
2. A snoop method.
3. A container method.
4. A listener method.

13. The fundamental classes for GUI programming are contained in the:

1. Java Enhancement Toolkit

2. Abstract Windowing Toolkit

3. Graphics Event Toolkit
4. GUI toolkit

14. Which of the following statements are true?

1. The collection interface contains a method called enumerator.
2. The iterator method returns an instance of the Vector class.

3. The Set interface is designed for unique elements.

4. At the root of the collection hierarchy is a class called Collection.

15. How do you indicate where a component will be positioned using `FlowLayout`?

1. North, South, East, West
2. Assign a row/column grid reference
3. Do nothing; the `FlowLayout` will position the component
4. Pass an X/Y percentage parameter to the add method

16. Why does an applet have no `main()` method?

1. The `paint()` method is like the main method for an applet.
2. The browser acts as the main. The applet provides methods for the browser.

3. Programs that do graphics don't need a main.

4. Only simple programs need a main.

17. What is the role of the Graphics object?

1. It contains a list of commands from the browser to your applet.

2. It can be omitted, but is sometimes used for file i/o.

3. It represents the area the applet can draw in and provides drawing methods.

4. It represents the entire screen of the computer monitor.

18. Which of the following puts "Hello" starting at X=20 Y=50?

1. graph.drawString(20, 50, "Hello");

2. drawString("Hello", 20, 50);

3. graph.drawString("Hello", 20, 50);

4. graph.println("Hello");

19. Which of the following draws a circle with a radius of 50 at X=10, Y=20?

1. graph.drawCircle(50, 50, 10, 20);

2. graph.drawCircle(50);

3. graph.drawOval(50, 50);

4. graph.drawOval(10, 20, 50, 50);

20. What must be imported to get class Applet?

1. java.awt.*

2. java.java

3. java.applet.Applet

4. java.util

21. What must be imported to get most graphics components?

1. java.*

2. java.util.*

3. java.Graphics

4. java.awt.*

22. What method call is used to draw a line from X=20, Y=30 to X=70, Y=50?

1. gr.drawLine(20, 30, 70, 50);

2. gr.drawLine((20, 30), (70, 50));

3. gr.drawLine(20, 70, 30, 50);

4. gr.drawLine(20, 30, 50, 70);

23. What method of your applet is called by the browser when it wishes to display it on the monitor?

1. setColor()

2. setBackground()

3. drawLine()

4. paint()

24. Can the source code for your applet be compiled by the usual javac compiler?

1. Yes—if you are going to run it from the DOS prompt.

2. Yes—an applet is just another class as far as the compiler is concerned.

3. No—because applets have no main()

4. No—the web browser compiles the code.

25. Can two or more applets appear on the same web page?

1. No, applets can't appear on web pages.
2. Yes, but there must be `<applet>` and `</applet>` for each one.
3. Yes, put them all between `<applet>` and `</applet>`
4. No, only one per page.

26. The syntax of `drawString()` method is:

1. `void drawstring(String x, int x)`
2. `void drawstring(String s)`
3. `void drawstring(int x, int y, string s)`
4. `void drawstring(String s, int x, int y)`

27. The syntax of `paint()` method is:

1. `public void paint(Graphics g)`
2. `public void paint(String s)`
3. `public void paint(Graphics g, String s)`
4. `public void paint()`

28. The constructor for `Font` class is:

1. `Font(String name, int style)`
2. `Font(String name, int style, int size)`
3. `Font(String name)`
4. `Font(String name, int size)`

29. To display text in the applet status bar which method is used?

1. `showStatus()`
2. `drawStatusBar()`
3. `drawStatus()`
4. `showStatusBar()`

30. Which of the following is an optional attribute of applet tag?

1. Code
2. Name
3. Height
4. Width

31. Which method is called first by an applet?

1. `run()`
2. `init()`
3. `paint()`
4. `start()`

32. What should be done to invoke the `run()` method on a thread for an object derived from the `Thread` class?

1. The `start()` method should be directly invoked on the Object.
2. The `init()` method should be directly invoked on the Object.
3. The `run()` method should be directly invoked on the Object.

4. The creation of the object using the new operator would create a new thread and invoke its run() method.

33. What is the default priority of a newly created thread?

1. MAX_PRIORITY (10)
2. A thread inherits the priority of its parent thread.
3. NORM_PRIORITY (5)
4. MIN_PRIORITY (1)

34. The class Hashtable is used to implement which collection interface?

1. SortedSet
2. List
3. Map
4. Set

35. Which of the following statements are true?

1. The notify() and notifyAll() methods can be used to signal and move waiting threads to ready-to-run state.
2. The wait(), notify(), and notifyAll() methods must be executed in synchronized code.
3. The Thread class is an abstract class.
4. The wait method defined in the Thread class can be used to convert a thread from Running to Waiting.

36. What is the name of the Collection interface used to represent elements in a sequence (in a particular order)?

1. Map
2. Set
3. List
4. Collection

37. Which of these classes implement the Collection interface SortedMap?

1. TreeMap
2. Hashtable
3. HashMap
4. TreeSet

38. The start() method invokes the run() method when the thread is ready to execute. What is the output?

1. Output of first0, first1, second0, second1
2. Compile time error
3. Output of first0, second0, first0, second1
4. Runtime error

39. Which most closely matches a description of a Java Map?

1. A class for containing unique array elements
2. A vector of arrays for a 2D geographic representation

3. A class for containing unique vector elements

4. An interface that ensures that implementing classes cannot contain duplicate keys

40. How does the set collection deal with duplicate elements?

1. Duplicate values will cause an error at compile time

2. An exception is thrown if you attempt to add an element with a duplicate value

3. The add method returns false if you attempt to add an element with a duplicate value

4. A set may contain elements that return duplicate values from a call to equals

41. What can cause a thread to stop executing?

1. A call to the thread's stop method.

2. Another thread is given a higher priority

3. The program exits via a call to `System.exit(0);`

4. A call to the halt method of the Thread class

42. Under what circumstances might you use the yield method of the Thread class?

1. To call from the currently running thread with a parameter designating which thread should run

2. To call from the currently running thread to allow another thread of the same or higher priority to run

3. To allow a thread of higher priority to run

4. To call on a waiting thread to allow it to run

43. Which of the following are methods of the Thread class?

1. go()

2. stop()

3. yield()

4. sleep(long msec)

44. Which of the following methods are members of the Vector class and allow you to input a new element?

1. addItem

2. append

3. addElement

4. insert

45. Which of the following are methods of the Collection interface?

1. isEmpty

2. iterator

3. setText

4. toArray

46. SortedMap is a/an?

1. Class

2. Interface

3. Method

4. Abstract Class

47. How can you prevent a member variable from becoming serialized?

1. You cannot.

2. By marking it transient

3. By marking it volatile

4. By marking it private

48. What can cause a thread to become non-runnable?

1. Calling the notifyAll method on an object.

2. Calling the notify method on an object.

3. Exiting from a synchronized block.

4. Calling the wait method on an object.

49. Declare and construct a Vector with 20 initial slots and an increment size of 5.

1. Vector v = new Vector(20)

2. Vector v(20) = new Vector(5)

3. Vector v = new Vector(20, 5)

4. Vector[] v = new Vector(20,5)

50. A Vector object acts like:

1. An array of primitive values.

2. An array of primitive values or of object references.

3. An array of references to objects of a particular class.

4. An array of references to objects of class Object.

51. Which of the following opens the file "myData.stuff" for Input?

1. FileInputStream fis = new
FileInputStream(new
BufferedInputStream("myData.stuff"
))

2. FileInputStream fis = new
FileInputStream("myData.stuff",
true)

3. FileInputStream fis = new
FileInputStream("myData.stuff")

4. DataInputStream dis = new
DataInputStream("myData.stuff")

52. Which of the following best describes the use of the synchronized keyword?

1. Ensures only one thread at a time
may access a method or object

2. Allows two processes to run in
parallel but communicate with each
other

3. Ensures that two or more Threads
will start and end at the same time

4. Ensures that two or more processes
will start and end at the same time

53. What package holds the File class?

1. java.file

2. java.io

3. java.lang

4. java.util

54. What method of File is used to test if a file or directory exists?

1. list()

2. isFile()

3. exists()

4. isDirectory()

55. A program uses the FileWriter constructor with the string "newFile.txt". What happens if "newFile.txt" already exists?

1. The constructor returns null.

2. The program exits.

3. The existing file is erased and replaced with a new, empty one.

4. An exception is thrown.

56. What File method creates a new disk directory?

1. dir()

2. mkdir()

3. list()

4. makeDirectory()

57. In the following, what is the directory separator character?

C:\MyFiles\Programs\Examples\someFile.txt

1. .

2. :

3. \

4. /

58. How many bits per character does a Java program use?

1. 16

2. 4

3. 32

4. 8

59. What does the flush() method of a stream do?

1. It ensures that all pending output operations are completed.

2. It erases all the data in the file.

3. It summons the garbage collector.

4. It initializes the output methods.

60. What data type does readLine() return?

1. String

2. int

3. char

4. byte[]

61. What class does readLine() belong to?

1. InputStreamReader

2. BufferedReader

3. FileReader

4. Reader

62. What abstract data type is the ancestor of all streams that output byte-oriented data?

1. **OutputStream**
2. **BinaryOutputStream**
3. **ByteStream**
4. **ByteOutputStream**

63. What is the `DataInputStream` method that reads an int value?

1. `ReadInt()`
2. **`readInt()`**
3. `read()`
4. `ReadInteger()`

64. What happens when the constructor for `FileInputStream` fails to open a file for reading?

1. It returns null.
2. It throws a `DataFormatException`.
3. It throws an `ArrayIndexOutOfBoundsException`.
4. **It throws a `FileNotFoundException`.**

65. Which of the following statements are true?

1. The `wait` method defined in the `Thread` class can be used to convert a thread from `Running` state to `Waiting`.
2. The `Thread` class is an abstract class.

3. **The `notify()` and `notifyAll()` methods can be used to signal and move waiting threads to ready-to-run state.**

4. **The `wait()`, `notify()`, and `notifyAll()` methods must be executed in synchronized code.**

66. Which of these classes implement the `Collection` interface `SortedMap`?

1. `Hashtable`
2. `HashMap`
3. `TreeSet`
4. **`TreeMap`**

67. How does the set collection deal with duplicate elements?

1. A set may contain elements that return duplicate values from `equals()`
2. **The `add` method returns false if you attempt to add an element with a duplicate value**
3. Duplicate values will cause an error at compile time
4. An exception is thrown if you attempt to add an element with a duplicate value

68. What can cause a thread to stop executing?

1. Another thread is given a higher priority
2. **A call to the thread's `stop` method.**

3. A call to the halt method of the Thread class

4. The program exits via a call to `System.exit(0);`

69. Which of the following are methods of the Thread class?

1. `sleep(long msec)`

2. `yield()`

3. `stop()`

4. `go()`

70. Which of the following are methods of the Collection interface?

1. `setText`

2. `iterator`

3. `toArray`

4. `isEmpty`

71. What is the name of the printed output of a program that shows the byte-by-byte contents of a binary file?

1. Pattern Listing

2. Binary Refuse

3. Hex Dump

4. Charmed Display